

PHYS 540 – Spring 2026

Computational Accelerator Physics

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Class: Tue/Thu 11:25-12:40 in PS 116

Office Hours: Tue/Thu 10:00-11:00, 12:45-13:45,

Fri 12:45-13:45, and by appointment

Course Outline

Single particle dynamics and numerical integration; transverse and longitudinal motion; phase space distributions and ellipses; transfer map methods; periodic systems; advanced beam optics modules; magnetic fields and FEM; RF cavities; errors and resonances; symplectic integration; other topics (final presentations).

Learning Objectives

- Build upon earlier material: E&M (PHYS 413) and relativity (PHYS 403).
- Math/physics connection: understand the basic underlying principles guiding charged particles (electric and magnetic fields, equations for transverse and longitudinal dynamics).
- Software: familiarize yourself with different accelerator simulation codes, understand the basics of their scripting language.
- Problem-solving techniques: be able to choose and apply the problem-solving technique that is appropriate for a particular problem; be able to apply these problem-solving approaches to novel contexts (i.e., to solve problems that do not map directly to examples covered in class).
- Problem-solving strategy: be able to draw upon an organized set of content knowledge and apply problem-solving techniques to that knowledge in order to organize and carry out long analyses of physical problems; be able to connect all the pieces of a problem to reach the final solution; recognize the value for learning the material of taking wrong turns; be able to recover from mistakes, and persist in working towards a solution.
- Communication: be able to justify and explain an approach to a problem in written or oral form (final project presentation and report).

Textbooks

Multiple recommended sources, no single required textbook

- The best source of the most up-to-date information is the list of U.S. Particle Accelerator School courses that can be found at <https://uspas.fnal.gov/materials/materials-table.shtml>
- We will be starting by using MAD-X (for a number of reasons), and its documentation is a good source of information, see here: <http://madx.web.cern.ch/madx/>
- D. A. Edwards, M. J. Syphers, *An Introduction to the Physics of High Energy Accelerators*, Wiley-VCH, 1st Ed. (1993), ISBN-10: 0471551638, ISBN-13: 978-0471551638

– Augmented by <https://sites.google.com/site/pohea16/>

- S. Y. Lee, *Accelerator Physics*, Fourth Ed., World Scientific (2018), ISBN-10: 9813274786, ISBN-13: 978-9813274785 (I have a Third Ed.)
- A. Chao, *Handbook of Accelerator Physics and Engineering*, 2nd Ed., Word Scientific (2013), ISBN-10: 9814417173, ISBN-13: 978-9814417174
- M. Berz, K. Makino, W. Wan, *An Introduction to Beam Physics*, 1st Ed., CRC Press (2014), ISBN-10: 9780750302630, ISBN-13: 978-0750302630
- M. Berz, *Modern Map Methods in Particle Beam Physics*, Volume 108, 1st Ed., Academic Press (1999), ISBN-10: 0120147505, ISBN-13: 978-0120147502

Project reports

L^AT_EX or LyX with T_EX source provided if possible, or Word/Open Office (if you can make it consistent). Overleaf (<https://www.overleaf.com/>) is highly recommended.

Approximately once every three weeks, 5 assignments, 15% of the final grade each. Final project of your choice presented in class during the finals weeks (May 5–10, 2025), 25% of the final grade.

Grading

- Technical accuracy (55%)
 - Are the results of the various exercises correct?
 - Is the discussion of the computational techniques accurate and correct?
- Supporting materials (15%)
 - Are the graphs, figures and tables clear, properly labeled and in the appropriate locations in the report?
 - Are the computer programs in the appendices of the report correctly referred to in the report?
 - Are the computer programs formatted and documented adequately, i.e. are there sufficient comment statements in the programs to allow another user to understand the code?
- Presentation of physics applications (15%)
 - Are the discussions, presentations and treatment of the physics applications technically and physically correct?
- Quality of writing (15%)
 - Does the report sound professional and well written?
 - Is the language clear and concise?
 - Are verbs in the appropriate tense?
 - Are punctuation and paragraphing correct?

Important Dates

Last day to drop with 100% tuition refund: Jan 20.

Spring break week begins: Mar 16.

Last day to withdraw: Mar 30.

Last day of classes: May 2.

Last day to request an incomplete grade: May 3.

Final exam week: May 4-9.

Math and Science Help

The Math and Science Help (MASH) under the Academic Resource Center provides academic support for undergraduate students through one-to-one peer-tutoring and review sessions via supplemental instruction. The program focuses on strengthening technical understanding, improving course performance, and developing effective study strategies that support long-term academic success.

Appointments are available five days a week during the fall and spring semesters. In-person meetings and walk-ins take place on the upper level of the Galvin Library, around the corner from OTS. While virtual appointments are available, it is highly encouraged to meet in person for a more effective learning experience. Visit illinoistech.mywconline.com to book an appointment with our tutors and supplemental instructors.

Communications Lab

The Illinois Tech Communications Lab offers in-person and online tutoring for undergraduate and graduate students. The tutors can assist with all written and oral assignments during any stage of the process, from brainstorming and outlining to final touches, citations, and delivery. Visit <https://www.iit.edu/hass/student-resources/communications-lab> for information about appointments, location, services, and more.

Academic Honesty

All students are expected to follow IIT's Academic Honesty policy which is found at <https://web.iit.edu/student-affairs/handbook/fine-print/code-academic-honesty>. The first violation results in a grade of zero for that assignment or exam. Subsequent violations result in a failing grade in the course.

Accommodations

Reasonable accommodation will be made for students with documented disabilities. In order to receive accommodations, students must obtain a letter of accommodation from the Center for Disability Resources as soon as possible. The Center for Disability Resources (CDR) is located in 3F3-1 at 10 W 35th St., telephone 312-567-5744 or email disabilities@iit.edu.

Illinois Tech's Sexual Harassment and Discrimination Information

Illinois Tech prohibits all sexual harassment, sexual misconduct, and gender discrimination by any member of our community. This includes harassment among students, staff, or faculty. Sexual harassment of a student by a faculty member or sexual harassment of an employee by a supervisor is particularly serious. Such conduct may easily create an intimidating, hostile, or offensive environment.

Illinois Tech encourages anyone experiencing sexual harassment or sexual misconduct to speak with the Office of Title IX Compliance for information on support options and the resolution process.

You can report sexual harassment electronically at iit.edu/incidentreport, which may be completed anonymously. You may additionally report by contacting the Title IX Coordinator, Virginia Foster at foster@iit.edu or the Deputy Title IX Coordinator at eespeland@iit.edu.

For confidential support, you may reach Illinois Tech's Confidential Advisor at (773) 907-1062. You can also contact a licensed practitioner in Illinois Tech's Student Health and Wellness Center at student.health@iit.edu or (312) 567-7550.

For a comprehensive list of resources regarding counseling services, medical assistance, legal assistance and visa and immigration services, you can visit the Office of Title IX Compliance website at <https://www.iit.edu/title-ix/resources>.